

Quantitative Thinking #3

Why Do Aircraft Wings Produce Lift?

Date

August 2026

Project Goal

I wanted to understand how aircraft wings generate lift and why airspeed plays such an important role in flight.

What I Worked On

- Researched the basic principles of aerodynamics.
- Learned how lift force is generated during flight.
- Studied the relationship between airspeed and lift.
- Explored the standard lift equation and its variables.
- Created an Excel spreadsheet to compare lift force at different airspeeds.
- Calculated lift force using the same wing area, air density, and lift coefficient.
- Organized the results into a data table.
- Created a graph to visualize how airspeed affects lift.

What I Learned

Before starting this analysis, I knew that faster aircraft generally generate more lift.

However, I did not realize how strongly lift depends on airspeed.

One of the most interesting discoveries was that lift does not increase at the same rate as speed. Even a relatively small increase in airspeed produced a much larger increase in lift force.

This analysis helped me better understand one of the basic ideas behind aerodynamics and aircraft flight.

Results

- Lift force increased as airspeed increased.
- The graph showed a clear upward curve rather than a straight line.
- Small increases in speed produced much larger increases in lift.
- The highest lift force occurred at the highest airspeed tested.

- The results showed how important airspeed is in generating lift.

Challenges

One of the most interesting parts of this project was understanding why lift increased so quickly as speed increased.

At first, I expected lift to increase at a similar rate to speed. However, the calculations showed that lift grows much faster because velocity is squared in the lift equation.

Key Lessons Learned

- Airspeed is one of the most important factors affecting lift.
- Lift does not increase at the same rate as speed.
- Small changes in speed can have a large effect on lift.
- Mathematics can be used to better understand flight.
- Graphs and data can help explain engineering concepts.

Final Reflection

This analysis started with a simple question: Why do aircraft wings produce more lift as speed increases?

Although the calculations were straightforward, the results helped me better understand one of the most important concepts in aerodynamics.

What interested me most was seeing how a relatively small increase in speed could produce a much larger increase in lift force. This project showed me how mathematics can be used to explain aircraft performance and how engineering concepts can be explored using simple models and data analysis.

A small increase in airspeed can produce a much larger increase in lift.